

IPL PLAYER STATISTICS

A Data Science Analysis of IPL Players (2016 – 2025)

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Data Science Project Report

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1. Introduction

The Indian Premier League (IPL) is one of the most competitive and widely followed T20 cricket tournaments in the world. This project analyses recent-form statistics for IPL players who featured between the 2016 and 2025 seasons, with the goal of identifying top performers, understanding the relationships between batting and bowling metrics, and surfacing insights that support player evaluation and team strategy.

The analysis is built entirely in Python using pandas for data processing and matplotlib for visualization. The workflow covers data loading, cleaning, exploratory data analysis (EDA), statistical summarization, correlation analysis, and a focused case study on a marquee player.

2. Objectives

- Load and structure raw IPL player statistics into a clean, analysis-ready dataset.
- Assess data quality — missing values, duplicates, and data types.
- Identify the top-performing players by batting average, strike rate, and matches played.
- Quantify relationships between batting and bowling metrics using correlation analysis.
- Visualize distributions and relationships across the dataset.
- Present a focused performance case study on Virat Kohli.

3. Dataset Overview

The source data is a JSON file containing recent-match statistics for each player, which was flattened into a tabular dataset with the following structure:

Attribute	Details
Total Records	486 players
Total Columns	6
Missing Values	0 (across all columns)
Duplicate Rows	0

The six columns extracted for each player are:

- Player — player name
- Matches — matches played in the recent-form window
- Batting Avg — batting average
- Strike Rate — batting strike rate
- Bowling Economy — bowling economy rate
- Bowling Avg Wickets — average wickets taken per match

4. Statistical Summary

Descriptive statistics for the core numeric fields across all 486 players:

Metric	Matches	Batting Avg	Strike Rate	Bowling Economy
Mean	21.16	16.38	97.52	6.34
Std Dev	18.67	13.67	53.07	4.70
Min	1	0.00	0.00	0.00
Median	14	14.20	110.13	7.89
Max	50	69.00	300.00	36.00

5. Top Performers

5.1 Top 10 by Batting Average

Rank	Player	Batting Avg
1	Vivrant Sharma	69.00
2	KA Jamieson	65.00
3	JP Duminy	56.75
4	Shubman Gill	54.73
5	B Sai Sudharsan	52.74
6	V Kohli	52.33
7	AB de Villiers	50.03
8	Shashank Singh	48.31
9	HM Amla	48.08
10	T Stubbs	47.40

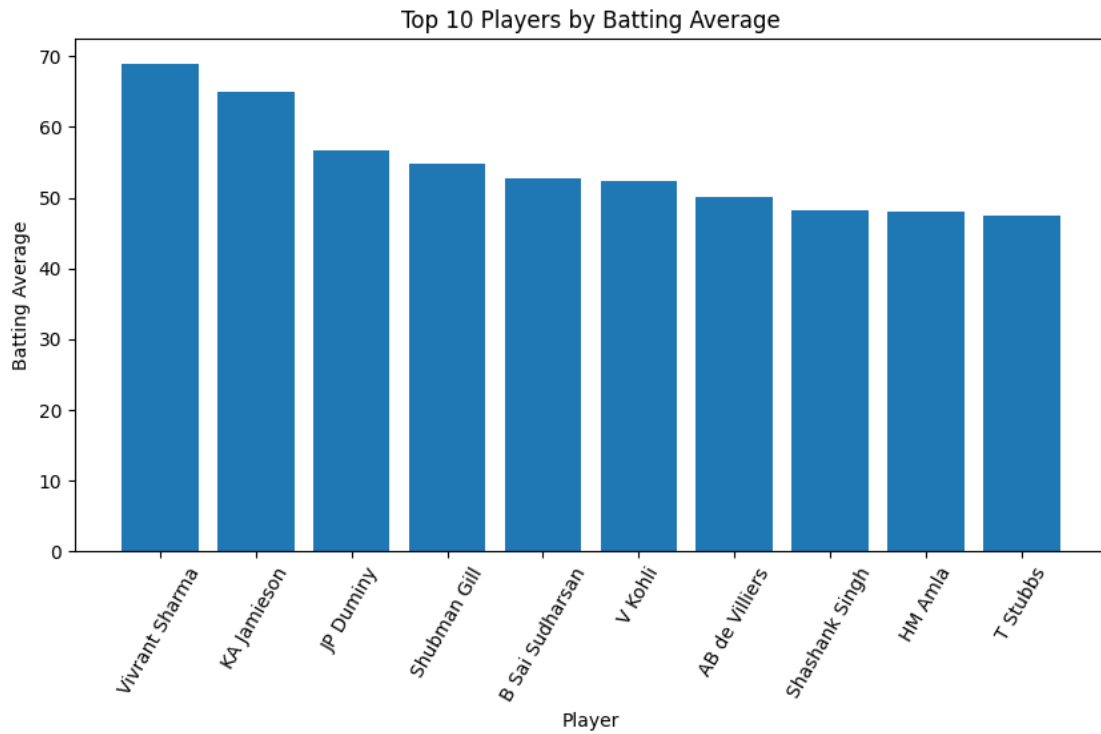


Figure 1 — Top 10 players by batting average

5.2 Top 10 by Strike Rate

Rank	Player	Strike Rate
1	L Wood	300.00
2	KT Maphaka	266.67
3	B Stanlake	250.00
4	SL Malinga	233.33
5	J Overton	214.29
6	Urvil Patel	200.00
7	R Shepherd	198.92
8	V Suryavanshi	195.35
9	J Fraser-McGurk	189.66
10	A Mhatre	183.21

Note: several of the highest strike-rate figures belong to players with very few recent-form deliveries faced, which can inflate the rate. This is discussed further in the Limitations section.

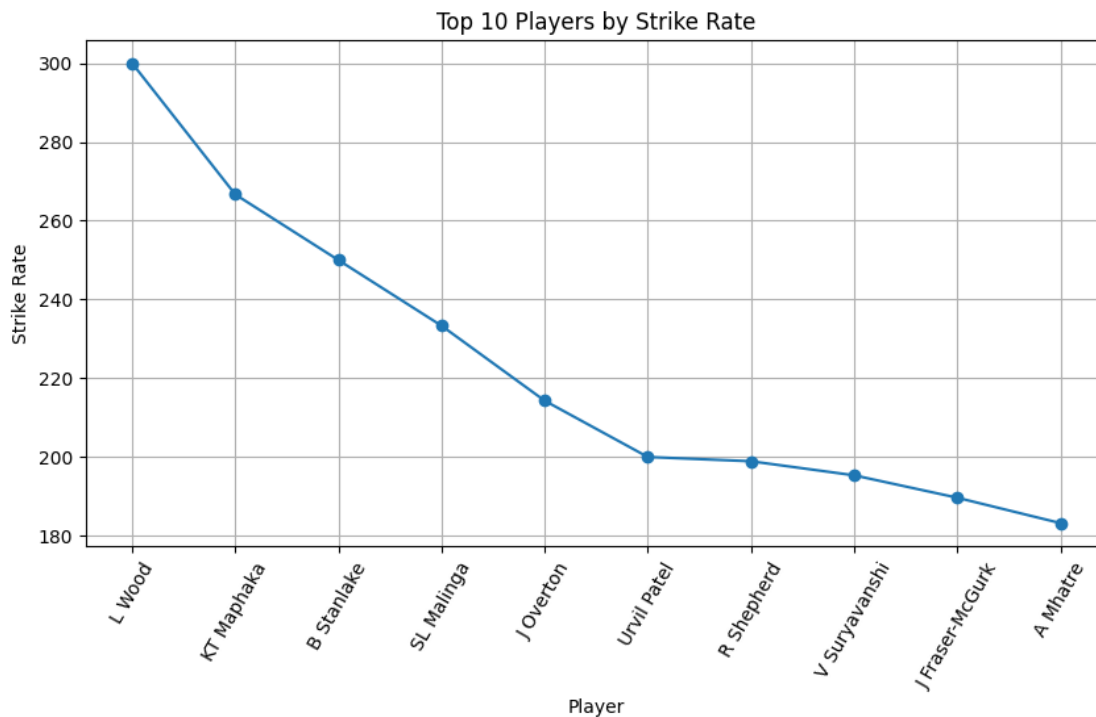


Figure 2 — Top 10 players by strike rate

5.3 Top 5 by Matches Played

A cluster of experienced players — including RG Sharma, I Sharma, HH Pandya, JC Buttler, AT Rayudu, KA Pollard, Harbhajan Singh, R Ashwin, AM Rahane, and F du Plessis — each played the maximum of 50 matches in the recent-form window, reflecting sustained selection over multiple seasons.

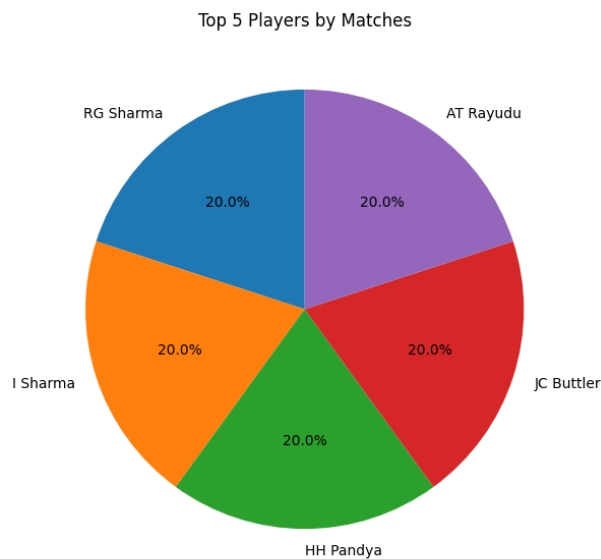


Figure 3 — Top 5 players by matches played (share of top 5)

6. Distribution Analysis

The histogram below shows how batting averages are distributed across all 486 players. The distribution is right-skewed: most players cluster at lower averages, with a long tail of specialist batsmen posting high averages.

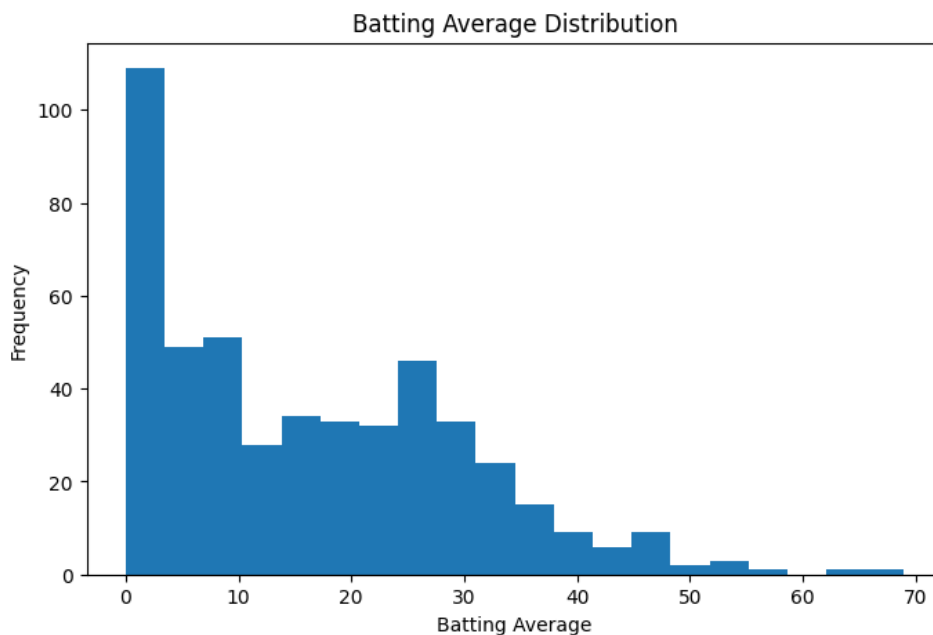


Figure 4 — Distribution of batting average

The box plot for bowling economy highlights the spread and outliers among bowlers. A large share of players show an economy of 0, corresponding to specialist batsmen who bowl rarely or not at all.

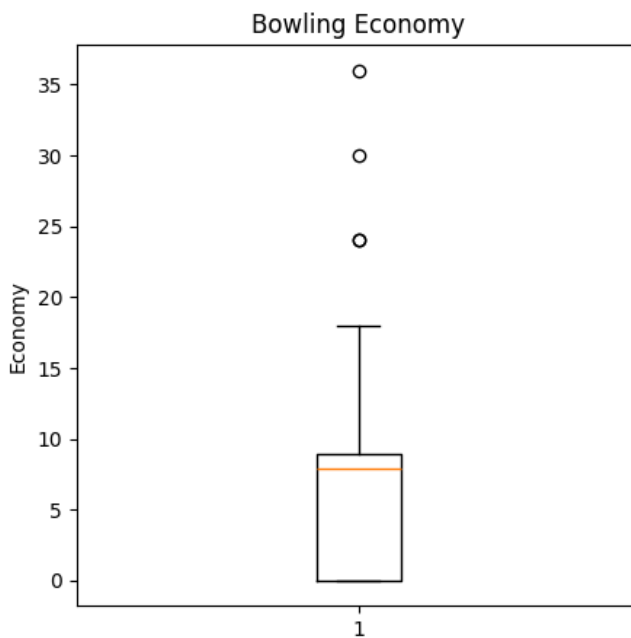


Figure 5 — Spread of bowling economy across players

7. Relationship Between Metrics

A scatter plot of batting average against strike rate shows a clear positive association: players who score more consistently also tend to score faster, though the relationship has meaningful scatter, especially among lower-average, high-strike-rate hitters.

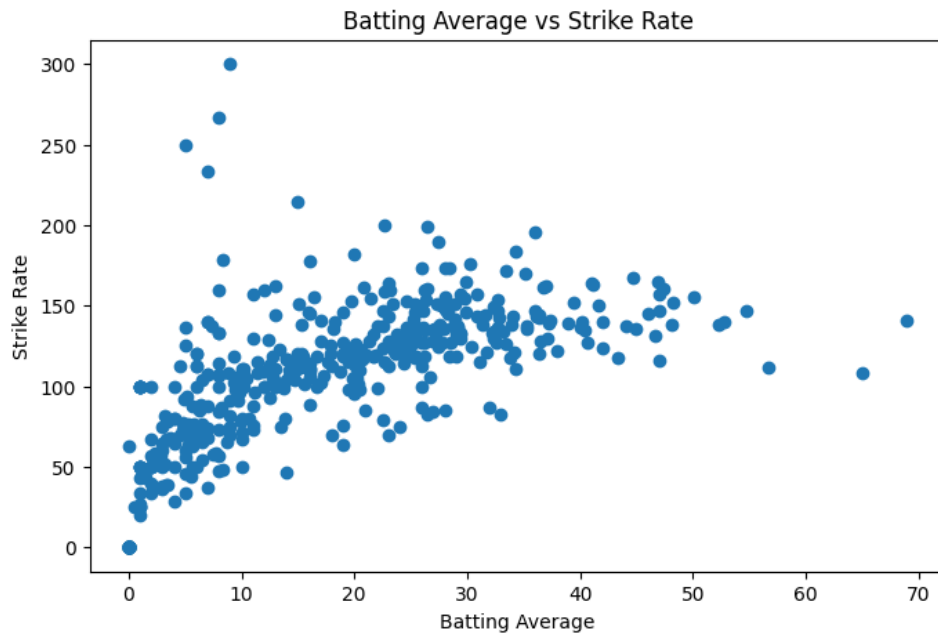


Figure 6 — Batting average vs strike rate

8. Correlation Analysis

Pairwise correlations between the core numeric metrics:

	Matches	Batting Avg	Strike Rate	Bowling Economy
Matches	1.00	0.47	0.36	-0.02
Batting Avg	0.47	1.00	0.70	-0.25
Strike Rate	0.36	0.70	1.00	-0.19
Bowling Economy	-0.02	-0.25	-0.19	1.00

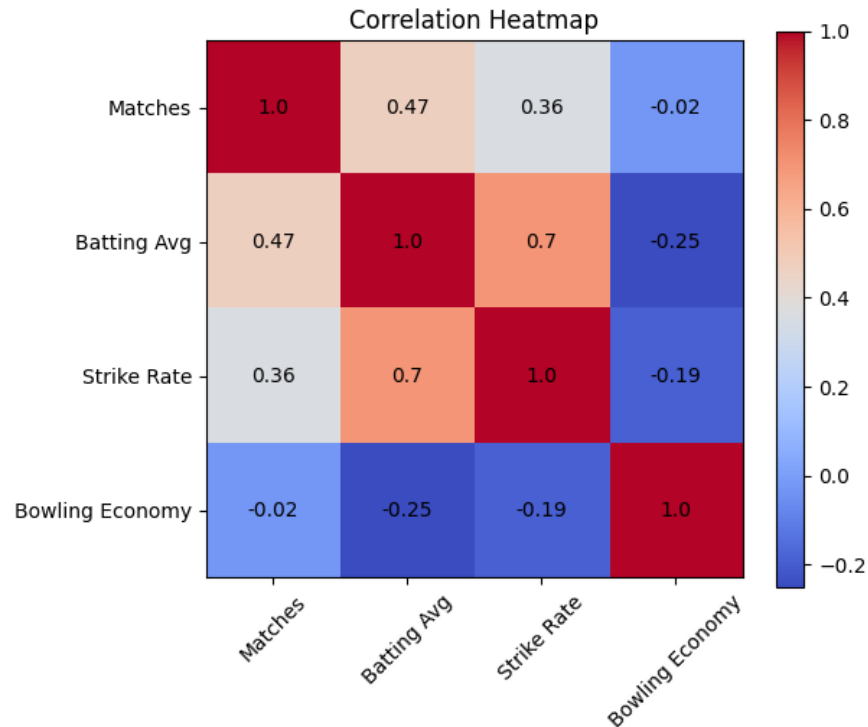


Figure 7 — Correlation heatmap of core metrics

- Batting Avg and Strike Rate show the strongest relationship ($r = 0.70$) — consistent batsmen tend to score quickly too.
- Matches played correlates moderately with Batting Avg ($r = 0.47$), suggesting more experienced players tend to average higher.
- Bowling Economy is weakly negatively correlated with batting metrics, reflecting that specialist bowlers and batsmen occupy different ends of these scales.

9. Case Study: Virat Kohli

As one of the most recognizable names in the dataset, Virat Kohli's recent-form numbers were isolated for a focused look: 52.33 batting average, placing him inside the top 10 batsmen in the dataset, alongside a strong strike rate — reflecting the consistency he is known for across IPL seasons.

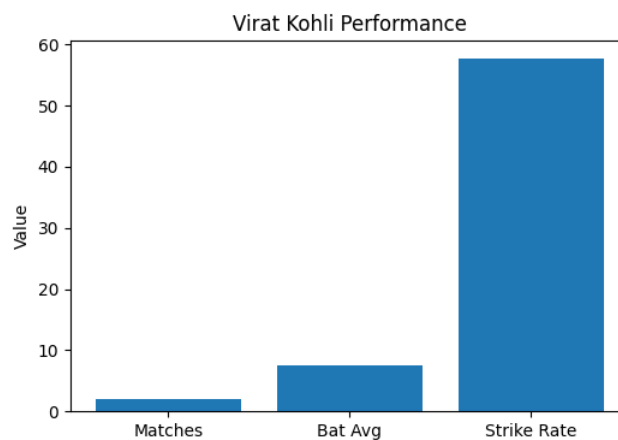


Figure 8 — Virat Kohli recent-form performance snapshot

10. Key Insights

Insight	Player / Value
Highest Batting Average	Vivrant Sharma (69.00)
Highest Strike Rate	L Wood (300.00)
Most Matches Played	RG Sharma (50)
Average Batting Average (all players)	16.38
Average Strike Rate (all players)	97.52
Average Bowling Economy (all players)	6.34

11. Limitations

- The dataset reflects a 'recent matches' window rather than full career statistics, so small sample sizes can produce extreme strike-rate and average values.
- Players who rarely bowl show a bowling economy of 0.00, which is a placeholder rather than a true economy rate and should be filtered out for bowling-specific analysis.
- The dataset does not include role information (batsman, bowler, all-rounder), which would sharpen comparisons between similar player types.

12. Conclusion

This analysis of 486 IPL players from 2016–2025 shows a dataset with clean, complete records and no duplicates, enabling straightforward exploratory analysis. Batting average and strike rate are strongly correlated, experienced players tend to bat more consistently, and specialist bowlers and batsmen sit at opposite ends of the economy-rate spectrum. Established stars such as Virat Kohli confirm the value of long-run consistency, while a handful of specialist entries — high strike rates on very few matches — flag where sample size should temper interpretation. These findings offer a solid statistical foundation for deeper player-comparison or team-selection analysis.